

LESSON 06 / 第六課

Tree Selection as a Trait Filter: From Regional Pools to the Planting Specification

作為性狀過濾的選樹：從區域庫到種植規格

Format. Short literature review. Claims that carry a number or a causal ranking are tied to a numbered source. Scale runs from the largest published pattern to the finest measured process. The lesson is not sequenced from the previous number and is not a regional case file unless the object itself is regional.

體例。短文獻綜述。帶數字或因果排序的陳述繫到編號文獻。尺度由最大已發表格局寫到最細已測過程。本課不由上一號遞推；除非對象本身是地域性的，否則不是單一地區檔案。

Abstract

摘要

Right tree, right place is a filter, not a slogan. At regional scale the pool is the climate and pest envelope plus whatever the nursery trade actually grows. At site scale the inventory is a set of trait constraints: mature crown and root volume, hydrology, load, targets, soils, utilities. The taxon comes last. Santamour's 10-20-30 statement is a heuristic ceiling against monodominance, not a measured optimum. Urban lists that collapse to a few dozen taxa fail that ceiling before a specialist pest arrives. A testable specification names the traits the site will enforce, then names a taxon that has been shown to meet them.

適樹適所是過濾器，不是口號。區域尺度上，庫是氣候與有害生物包絡，加上苗木貿易實際在種的東西。立地尺度上，清查是一套性狀約束：成熟樹冠與根體積、水文、荷載、目標、土壤、管線。分類群放在最後。Santamour 的 10-20-30 陳述是對抗單優的啟發式上限，不是測得的最適。塌縮到數十個分類群的城市名單，在專性有害生物到來之前就已未通過該上限。可檢驗的規格先寫明立地將執行的性狀，再寫明已被證明能滿足它們的分類群。

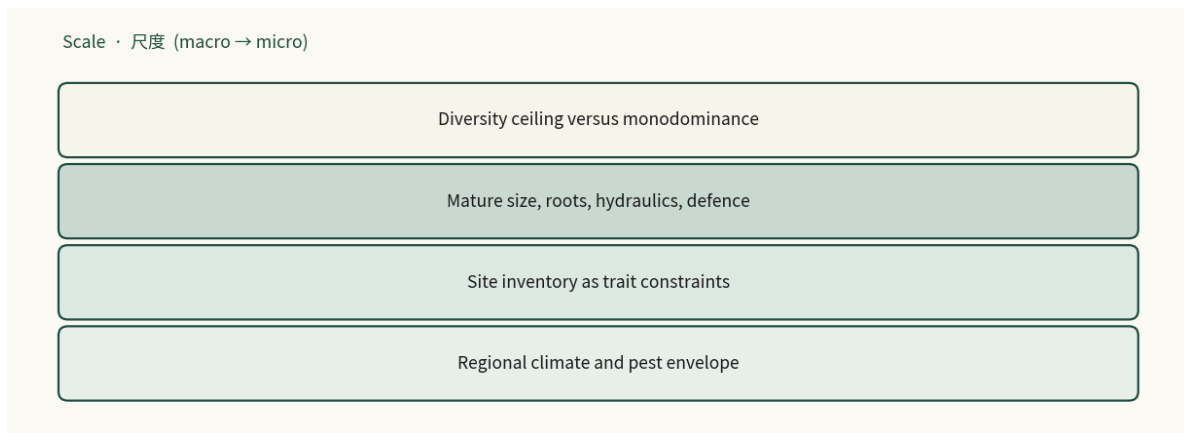


Figure 1. Nested scales from the regional pool to diversity rules.

圖1. 由區域庫到多樣性規則的嵌套尺度。

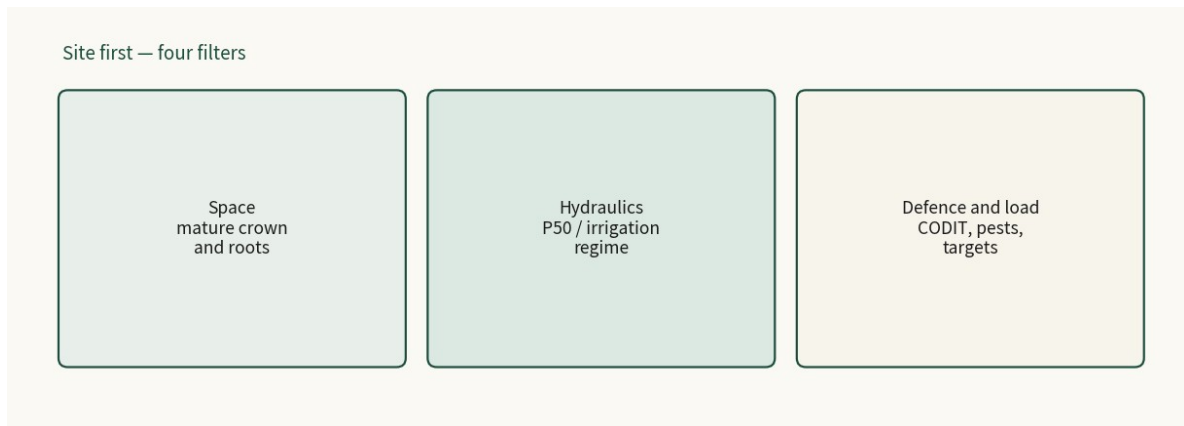


Figure 2. Site inventory first, taxon last.

圖2. 立地清查在先，分類群在後。

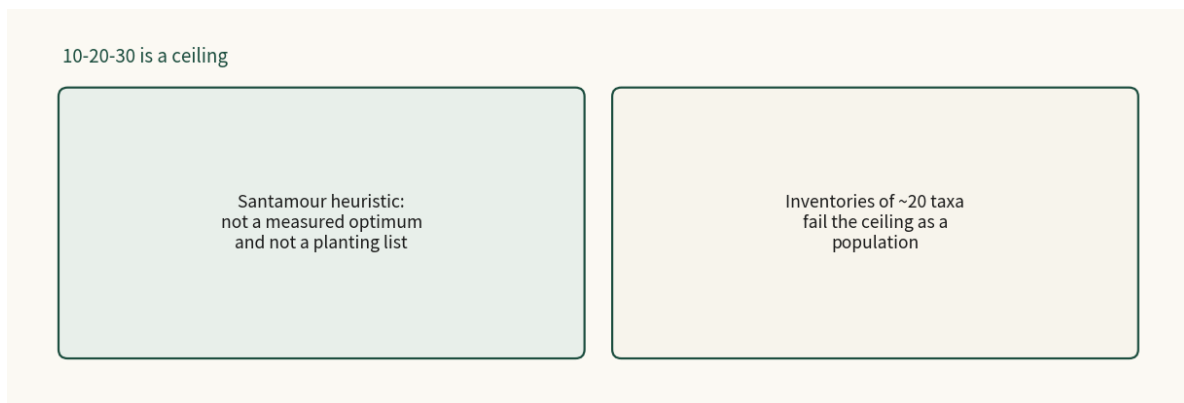


Figure 3. 10-20-30 is a heuristic ceiling, not a measured optimum.

圖3. 10-20-30 是啟發式上限，不是測得的最適。

1. Scope

1. 範圍

The object is published selection logic: how a site's constraints pick a taxon. Cultivar marketing is out of scope except where trials exist.

對象是已發表的選擇邏輯：立地約束如何挑選分類群。品種行銷不在範圍內，除非已有試驗。

2. Pool and filters

2. 庫與過濾

The pool is smaller than the flora. Climate envelope, pest guilds, nursery production and ordinance lists jointly decide what can be planted at scale [1,2]. Mature size and form, root behaviour, hydraulic strategy, defence, and the load/target field are separate filters [3,4]. A taxon that passes climate but fails underground utilities is not a pass. Filters are AND gates. The taxon is named last.

庫小於區系。氣候包絡、病蟲功能群、苗圃生產與法令名單共同決定什麼能被規模種植 [1,2]。成熟尺寸與形、根系行為、水力策略、防禦，以及荷載／目標場，是分開的過濾器 [3,4]。通過氣候但未通過地下管線的分類群，不是通過。過濾器是 AND 閘。分類群最後才命名。

3. Diversity ceilings and specification

3. 多樣性上限與規格

Santamour proposed that no more than 10% of a city's trees be of one species, 20% of one genus, 30% of one family — a heuristic against the next specialist outbreak [5]. It is not a measured optimum and it does not specify which 10%. Cities whose street inventories collapse to about twenty taxa fail the ceiling as a population [2,6]. A specification that names only a Latin binomial has not constrained mature height, root volume, salt, drought or a pest of record. Substitution rules belong in the same document.

Santamour 提出一城之樹同一種不超過 10%、同一屬 20%、同一科 30%——對抗下一次專性爆發的啟發式 [5]。它不是測得的最適，也不指定是哪 10%。行道清查塌縮到約二十個分類群的城市，作為族群已未通過上限 [2,6]。只寫拉丁雙名的規格，尚未約束成熟高度、根體積、鹽、乾旱或有紀錄的有害生物。替代規則屬於同一文件。

4. Limits

4. 限度

Trial data are richer in temperate North America and Europe than in tropical cities. Transferring a 'proven street tree' across biomes without a hydrology and pest check is a catalogue act, not selection.

試驗數據在溫帶北美與歐洲富於熱帶城市。沒有水文與有害生物核對就把「已驗證行道樹」跨生物群系轉移，是目錄行為，不是選擇。

Study questions

研習題

Choose the option that retains the published result. Answers follow the last item.

選出保留已發表結果的選項。答案在最後一題之後。

1. In this review the taxon is named:

1. 本綜述中分類群被命名於：

- A. First, before the site is seen.
- A. 最先，在看見立地之前。
- B. Last, after trait filters.
- B. 最後，在性狀過濾之後。
- C. Only by flower colour.
- C. 只靠花色。
- D. By t/R.
- D. 靠 t/R。

2. Filters in selection are:

2. 選擇中的過濾器是：

- A. OR gates.
- A. OR 閘。
- B. AND gates.
- B. AND 閘。
- C. Paint colours.
- C. 漆色。
- D. Flash densities.
- D. 閃擊密度。

3. Santamour 10-20-30 is:

3. Santamour 10-20-30 是：

- A. A measured global optimum.
A. 測得的全球最適。
- B. A heuristic ceiling against monodominance.
B. 對抗單優的啟發式上限。
- C. A P50 table.
C. P50 表。
- D. Wall 3.
D. 第 3 壁。

4. A list that collapses to ~20 urban taxa:

4. 塌縮到約 20 個城市分類群的名單：

- A. Passes the diversity ceiling automatically.
A. 自動通過多樣性上限。
- B. Fails the ceiling as a population.
B. 作為族群未通過上限。
- C. Raises safety margins.
C. 提高安全邊際。
- D. Is a tomogram.
D. 是斷層圖。

5. A binomial-only specification:

5. 只有雙名的規格：

- A. Constrains mature size and hydrology.
A. 約束了成熟尺寸與水文。
- B. Does not yet constrain those traits.
B. 尚未約束那些性狀。
- C. Is a vulnerability curve.
C. 是脆弱性曲線。
- D. Is IPM.
D. 是 IPM。

6. Substitution rules belong:

6. 替代規則屬於：

- A. Nowhere.
A. 無處。
- B. In the same specification document.
B. 同一規格文件。
- C. Only in folklore.
C. 只有傳聞。
- D. In lightning standards.
D. 防雷標準。

7. Nursery production constrains the pool because:

7. 苗圃生產約束庫，因為：

- A. Arboretum plants are always available at street scale.

- A. 樹木園植物總能以街道尺度取得。
- B. What is not in production is not a street option this year.
- B. 不在生產中的，今年就不是行道選項。
- C. All taxa are produced equally.
- C. 所有分類群被同等生產。
- D. P88 forbids nurseries.
- D. P88 禁止苗圃。

8. Hydraulic strategy as a filter asks:

8. 作為過濾器的水力策略問的是：

- A. Flower scent.
- A. 花香。
- B. Whether the taxon's P50 / phenology matches the site water regime.
- B. 該分類群的 P50／物候是否匹配立地水分體制。
- C. Only bark colour.
- C. 只有樹皮顏色。
- D. Only i-Tree.
- D. 只有 i-Tree。

9. 'Proven street tree' transferred across biomes without checks is:

9. 沒有核對就跨生物群系轉移的「已驗證行道樹」是：

- A. Selection.
- A. 選擇。
- B. A catalogue act.
- B. 目錄行為。
- C. CODIT Wall 4.
- C. CODIT 第 4 壁。
- D. A register diagnosis.
- D. 名錄診斷。

10. Site inventory in this review is:

10. 本綜述中的立地清查是：

- A. Optional decoration.
- A. 可選裝飾。
- B. The source of trait constraints.
- B. 性狀約束的來源。
- C. A decay fungus.
- C. 腐朽菌。
- D. A cable termination.
- D. 鋼纜錨頭。

Answers / 答案 1-B 2-B 3-B 4-B 5-B 6-B 7-B 8-B 9-B 10-B

References

參考文獻

[1] Sæbø A., Benedikz T., Randrup T.B. selection criteria for urban trees.

- [2] Richards N.A. diversity of the urban forest and street-tree lists.
- [3] Miller R.W., Hauer R.J., Werner L.P. Urban Forestry.
- [4] Sjöman H., Hirons A. trait-based selection and climate envelopes.
- [5] Santamour F.S. Jr. (1990). Metria 7.
- [6] Raupp M.J., Cumming A.B., Raupp E.C. street-tree diversity and pest vulnerability.
- [7] Clark J.R. et al. urban forest sustainability and species fitness to site.

Closing note. This lesson is written as a short review article. Disagreement in the literature is kept in view where the results actually diverge.

結語。本課以短綜述文章寫成。文獻真正分歧之處，兩造結果都保留在視野內。